Tarea No. 2.1

2.55

a) $X - X_0 = \left(\frac{V_{0x} + V_x}{2}\right) t$

$X - 0 = \left(\frac{0 + 10 m/s}{2}\right) 2s$

$X = \left(\frac{10}{2}\right) 2 = 10m$

b) $x = 50$ $\quad V = \frac{50}{6}$ $\quad V = 8.33 m/s$

$X = 100$ $\quad V = \frac{100}{11}$ $\quad V = 9.09 m/s$

$X = 200$ $\quad V = \frac{200}{21}$ $\quad V = 9.52 m/s$

2.59 Estudio de los terremotos

$t = \frac{d}{V}$ $P = 6.5 Km/s$
$S = 3.5 Km/s$

$\frac{d}{6.5} - \frac{d}{3.5} = 33 seg$ $d\left|\left(\frac{1}{6.5}\right) - \left(\frac{1}{3.5}\right)\right| = 33$

$d = \frac{33}{12/91} = \underline{250 Km}$

2.61

a) $\frac{X_2 - X_1}{t_2 - t_1} = \frac{1,000m - 63m}{4.75s} = \frac{937m}{4.75s} = \underline{197 m/s}$

b) $\frac{1000m - 0}{5.90s} = \frac{1000m}{5.9} = \underline{169.5 m/s}$

2.63

$S \rightarrow A$

$Vel = 88 Km/h$ $V = \frac{d}{t}$ $88 Km/h = \frac{76 Km}{t}$ $t = 0.86 h$
$D = 76 Km$
$t = ?$

A → y

Vel = 72 Km/h
D = 34 Km
t = ?

$t = \frac{34 Km}{72 Km/h} = 0.47 h$

a) $rap\ med = \frac{76 Km + 34 Km}{0.86 h + 0.47 h} = \frac{110}{1.33} = \underline{82.7 Km/h}$

b) $vel\ med = \frac{76 Km - 34 Km}{0.86 h + 0.47 h} = \frac{42}{1.33} = \underline{31 Km/h}$

2.65

$\Delta X = 100 m$ $\Delta t = (9.1 s) - (4.5)$ $V = \frac{\Delta X}{t} = \frac{100}{5.7} = \underline{19.61 m/s}$

a) $V_x = V_{0x} + a_x t$

$27.74 m/s = 0 + a_x (4 s)$
$a_x = \frac{27.74 m/s}{4 s} = \underline{3.5 m/s^2}$

b) en el segundo tramo $a = 0$

c) $X = X_0 + V_0 t + \frac{1}{2} a t^2$
$100 m = 0 + 0 + \frac{1}{2} a (9.1)^2$
$100 = 41.41 a$
$a = 100/41.41 = \underline{2.41 m/s^2}$

2.67

a) $x - x_0 = \left(\frac{V_{0x} + V_x}{2}\right) t = x - 0 = \left(\frac{4+12}{2}\right) 12$

$= x = (8) 12 = 96$

$96 - 4 = \underline{92m}$

b) $x = \left(12m + 4m/seg\right)\frac{10}{2} = 80m$

$x = 80 + 12 = \underline{92m}$

2.69

$150m = V_i \cdot 5s + \frac{a}{2} 25s^2$

$V_i = a5_s$

$V_i = 20m/s$ $a = 4m/seg^2$ $\frac{V_i^2}{2a} = \frac{400}{8} = \underline{50m}$

2.71)

$V_c = 1.5m/s$ } separación $= 0.90m$ $X_h = 1.2 + 0.9 = 2.1$
$V_h = 0.8m/s$ } $X = 1.2m$

$t = \frac{d}{V}$ $t = \frac{1.2}{1.5} = 0.8$

$X = X_0 + V_{0x} t + \frac{1}{2} a_x t^2$

$X_h = V_h + \frac{1}{2} a t^2$

$a = 2\left(\frac{X_h - V_h t}{t^2}\right) = a = 2\left(\frac{2.1\,m - (0.8\,m/s)(0.8)}{(0.8\,m/s)^2}\right)$

$a = \underline{4.6\,m/s^2}$

2.73

a) $X = X_0 + V_0 t + \frac{1}{2} a t^2$

$X = \frac{1}{2} a t^2 \quad t = \sqrt{\frac{2X}{a}} = \sqrt{\frac{2 * 40}{2.1}} = \underline{6.17\,s}$

b) $X = \frac{1}{2} a t^2 = 40 + d$

$d = \frac{1}{2}(3.4)(6.17)^2 - 40 = \underline{24.8}$

c) $V_x^2 = V_0^2 + 2a(X - X_0)$

$V_x = \sqrt{2aX'} = 12.96 \approx \underline{13\,m/s}$

$V_x = \sqrt{2aX'} = 20.99 \approx \underline{21\,m/s}$

2.75

b)

$$V_{med} = \frac{50\,cm - 0}{10\,s - 0} = \underline{5\,cm/s}$$

a) $$V_{med} = \frac{78\,cm - 0}{10\,s - 0} = \underline{7.8\,cm/s}$$

Scanned with CamScanner